

TEST PLAN DOCUMENT

Attendance Management System

1. Test Plan Overview

Purpose: To verify that the Attendance Management System satisfies all functional and non-functional requirements defined in the SRS, including authentication, attendance marking, record management, percentage calculation, attendance viewing, reporting, security, performance and reliability.

2. Requirements Covered

Requirement ID	Module / Category	Requirement Summary
FR-01	Authentication & Role Management	Authorized Administrator, Teacher and Student login according to assigned role.
FR-02	Attendance Marking	Teachers select class/subject and mark Present, Absent or Late for a date.
FR-03	Attendance Record Management	Save/update records and prevent duplicates for student, subject and date.
FR-04	Attendance Percentage Calculation	Automatically calculate and display attendance percentage.
FR-05	Attendance Viewing	Students and teachers view attendance by subject and date.
FR-06	Attendance Reporting	Generate reports by student, class, subject or date range.
NFR-01	Security	Role-based access and protection of sensitive data during transmission.
NFR-02	Performance	Major pages and attendance information load within 3 seconds.
NFR-03	Reliability	Preserve records without data loss and perform regular backups.

FR-01 – Authentication & Role Management

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_FR-01_01	Authentication & Role Management	Valid login	Valid account exists	Open login; enter credentials; submit	Teacher01 / valid password	Correct role dashboard opens.	Pending	Pending
UT_FR-01_02	Authentication & Role Management	Invalid password	Login page available	Enter wrong password; submit	Teacher01 / Wrong@123	Error shown; access denied.	Pending	Pending
UT_FR-01_03	Authentication & Role Management	Blank credentials	Login page available	Submit empty fields	Blank username/ password	Required-field messages shown.	Pending	Pending
IT_FR-01_04	Authentication & Role Management	Administrator role mapping	Admin account exists	Login as administrator	Admin account	Admin functions are available.	Pending	Pending
IT_FR-01_05	Authentication & Role Management	Teacher role mapping	Teacher account exists	Login as teacher	Teacher account	Teacher functions are available.	Pending	Pending
IT_FR-01_06	Authentication & Role Management	Student role mapping	Student account exists	Login as student	Student account	Student functions are available.	Pending	Pending
IT_FR-01_07	Authentication & Role Management	Unauthorized page access	Student logged in	Open teacher/ admin page	Student session	Access denied or redirected.	Pending	Pending
SYS_FR-01_08	Authentication & Role Management	Logout	User logged in	Click logout; press back	Teacher account	Protected pages cannot be reopened.	Pending	Pending
SYS_FR-01_09	Authentication & Role Management	Session timeout	Timeout configured	Remain inactive; open protected page	Idle session	Login is required again.	Pending	Pending
SYS_FR-01_10	Authentication & Role Management	Authentication failure handling	Application available	Attempt login while auth service unavailable	Valid credentials	Safe error shown without technical details.	Pending	Pending

FR-02 – Attendance Marking

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_FR-02_01	Attendance Marking	Mark Present	Teacher logged in; class exists	Select class/ subject/date; mark Present; save	S01, DBMS, 2026-09-01, Present	Student marked Present.	Pending	Pending
UT_FR-02_02	Attendance Marking	Mark Absent	Roster loaded	Mark Absent; save	S01, Absent	Student marked Absent.	Pending	Pending
UT_FR-02_03	Attendance Marking	Mark Late	Roster loaded	Mark Late; save	S01, Late	Student marked Late.	Pending	Pending
UT_FR-02_04	Attendance Marking	Missing date validation	Teacher logged in	Leave date empty; save	Date blank	Validation message displayed.	Pending	Pending
UT_FR-02_05	Attendance Marking	Missing status validation	Roster loaded	Leave status unselected; save	Status blank	Validation message displayed.	Pending	Pending
IT_FR-02_06	Attendance Marking	Load class roster	Class and subject data exist	Select class and subject	CSE-A, DBMS	Correct enrolled students appear.	Pending	Pending
IT_FR-02_07	Attendance Marking	Save complete roster	Roster loaded	Mark statuses for all students; save	S01 Present; S02 Absent; S03 Late	All entries are processed.	Pending	Pending
IT_FR-02_08	Attendance Marking	Unauthorized subject	Teacher logged in	Attempt unassigned subject	Unassigned subject	Subject unavailable or access denied.	Pending	Pending
SYS_FR-02_09	Attendance Marking	Change attendance date	Roster loaded	Select another date	2026-09-02	Records correspond to selected date.	Pending	Pending
SYS_FR-02_10	Attendance Marking	Invalid attendance status	Roster loaded	Submit unsupported status	Excused	Invalid status rejected.	Pending	Pending

FR-03 – Attendance Record Management

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_FR-03_01	Attendance Record Management	Save new record	Valid attendance form	Enter data; save	S01, DBMS, 2026-09-01, Present	Record saved.	Pending	Pending
UT_FR-03_02	Attendance Record Management	Update existing record	Record exists	Change Present to Late; save	Existing S01 record	Record updated.	Pending	Pending
UT_FR-03_03	Attendance Record Management	Prevent duplicate record	Record exists	Submit same student/ subject/date	Same key values	Duplicate prevented.	Pending	Pending
UT_FR-03_04	Attendance Record Management	Missing student validation	Form available	Submit without student	Student blank	Validation error displayed.	Pending	Pending
UT_FR-03_05	Attendance Record Management	Missing subject validation	Form available	Submit without subject	Subject blank	Validation error displayed.	Pending	Pending
IT_FR-03_06	Attendance Record Management	Database persistence	Database connected	Save; reload record list	Valid record	Record remains available.	Pending	Pending
IT_FR-03_07	Attendance Record Management	Student update restriction	Student account exists	Login as student; attempt update	Student session	Update denied.	Pending	Pending
IT_FR-03_08	Attendance Record Management	Teacher authorization	Teacher assigned to class	Update assigned record	Authorized teacher	Update succeeds.	Pending	Pending
SYS_FR-03_09	Attendance Record Management	Refresh after save	Record saved	Refresh page	Saved record	Record is retained once.	Pending	Pending
SYS_FR-03_10	Attendance Record Management	Database failure handling	Database unavailable	Attempt save	Valid attendance data	Safe error; no partial record.	Pending	Pending

FR-04 – Attendance Percentage Calculation

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_FR-04_01	Attendance Percentage Calculation	All present calculation	10 records exist	Open percentage	10 Present / 10 classes	100% displayed.	Pending	Pending
UT_FR-04_02	Attendance Percentage Calculation	Half attendance calculation	10 records exist	Open percentage	5 Present / 10 classes	50% displayed.	Pending	Pending
UT_FR-04_03	Attendance Percentage Calculation	Absent calculation	Records exist	Open percentage	6 Present / 4 Absent	60% displayed.	Pending	Pending
UT_FR-04_04	Attendance Percentage Calculation	Late policy calculation	Late rule configured	Open percentage	8 Present / 2 Late	Configured late policy applied.	Pending	Pending
UT_FR-04_05	Attendance Percentage Calculation	No records calculation	No records exist	Open percentage	0 records	0% or no-records message shown.	Pending	Pending
IT_FR-04_06	Attendance Percentage Calculation	Recalculate after update	Existing records exist	Change status; refresh	Present changed to Absent	Percentage updates correctly.	Pending	Pending
IT_FR-04_07	Attendance Percentage Calculation	Subject filter calculation	Multiple subjects exist	Select one subject	DBMS only	Only selected subject included.	Pending	Pending
IT_FR-04_08	Attendance Percentage Calculation	Date range calculation	Records across dates	Select date range	2026-09-01 to 2026-09-10	Only selected range included.	Pending	Pending
SYS_FR-04_09	Attendance Percentage Calculation	Rounding calculation	Repeating decimal possible	Open percentage	2 Present / 3 classes	Consistent rounded percentage shown.	Pending	Pending
SYS_FR-04_10	Attendance Percentage Calculation	Large dataset calculation	Large dataset exists	Open percentage	1000+ records	Correct result within performance target.	Pending	Pending

FR-05 – Attendance Viewing

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_FR-05_01	Attendance Viewing	View own records	Student logged in; records exist	Open attendance page	Student S01	Own records displayed.	Pending	Pending
UT_FR-05_02	Attendance Viewing	View by subject	Records exist	Apply subject filter	DBMS	Only DBMS records displayed.	Pending	Pending
UT_FR-05_03	Attendance Viewing	View by date	Records exist	Apply date filter	2026-09-01	Matching date displayed.	Pending	Pending
UT_FR-05_04	Attendance Viewing	Empty result handling	No matching records	Apply filter	Nonexistent date	Clear empty-state message shown.	Pending	Pending
IT_FR-05_05	Attendance Viewing	Teacher views assigned class	Teacher assigned to class	Open class records	CSE-A	Authorized records displayed.	Pending	Pending
IT_FR-05_06	Attendance Viewing	Student data restriction	Two student accounts exist	Attempt to view another student	S02 while logged as S01	Access denied.	Pending	Pending
IT_FR-05_07	Attendance Viewing	Updated record visibility	Record updated	Open record	S01 Late	Latest value displayed.	Pending	Pending
SYS_FR-05_08	Attendance Viewing	Sort records	Records exist	Sort by date and subject	Multiple records	Correct order displayed.	Pending	Pending
SYS_FR-05_09	Attendance Viewing	Pagination	Many records exist	Move between pages	100+ records	Correct records shown per page.	Pending	Pending
SYS_FR-05_10	Attendance Viewing	Responsive viewing	Supported browser/device	Open at different widths	Desktop/mobile widths	Content remains readable.	Pending	Pending

FR-06 – Attendance Reporting

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_FR-06_01	Attendance Reporting	Report by student	Records exist	Select student report	S01	Only S01 data included.	Pending	Pending
UT_FR-06_02	Attendance Reporting	Report by class	Records exist	Select class report	CSE-A	Selected class data included.	Pending	Pending
UT_FR-06_03	Attendance Reporting	Report by subject	Records exist	Select subject report	DBMS	Selected subject data included.	Pending	Pending
UT_FR-06_04	Attendance Reporting	Report by date range	Records exist	Select start/end dates	2026-09-01 to 2026-09-10	Only range included.	Pending	Pending
IT_FR-06_05	Attendance Reporting	Combined filters	Records exist	Select class, subject and range	CSE-A, DBMS, date range	All filters applied.	Pending	Pending
IT_FR-06_06	Attendance Reporting	Current saved data	Records saved	Update record; generate report	Updated S01 record	Report uses latest data.	Pending	Pending
IT_FR-06_07	Attendance Reporting	Report access control	Role accounts exist	Student requests class-wide report	Student account	Only permitted scope available.	Pending	Pending
SYS_FR-06_08	Attendance Reporting	Empty report	No matching data	Generate report	No matching filter	Empty-report message shown.	Pending	Pending
SYS_FR-06_09	Attendance Reporting	Report accuracy	Known dataset exists	Generate and compare totals	Known totals	Totals and percentages match records.	Pending	Pending
SYS_FR-06_10	Attendance Reporting	Report performance	Large dataset exists	Generate report	Large class/ date range	Report generated within 3 seconds.	Pending	Pending

NFR-01 – Security

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_NFR-01_01	Security	Password field masking	Login page available	Type password	Password value	Password is not displayed in plain text.	Pending	Pending
IT_NFR-01_02	Security	Role permission enforcement	Role accounts exist	Access functions for each role	Admin/Teacher/Student	Only permitted functions are accessible.	Pending	Pending
IT_NFR-01_03	Security	Record scope protection	Records exist	Request another user's record	Unauthorized student ID	Request denied.	Pending	Pending
SYS_NFR-01_04	Security	Secure transmission	HTTPS deployment available	Login and inspect connection	Credentials over HTTPS	Sensitive data transmitted securely.	Pending	Pending
SYS_NFR-01_05	Security	Session protection	User logged in	Copy protected URL to logged-out browser	Protected URL	Login required.	Pending	Pending

NFR-02 – Performance

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
SYS_NFR-02_01	Performance	Login response time	Application available	Login and measure response	Valid account	Login completes within 3 seconds.	Pending	Pending
SYS_NFR-02_02	Performance	Dashboard load time	User logged in	Open dashboard and measure	Teacher account	Dashboard loads within 3 seconds.	Pending	Pending
SYS_NFR-02_03	Performance	Attendance page load	Records exist	Open attendance page	Normal dataset	Page loads within 3 seconds.	Pending	Pending
SYS_NFR-02_04	Performance	Percentage calculation time	Records exist	Open summary and measure	Normal dataset	Percentage appears within 3 seconds.	Pending	Pending
SYS_NFR-02_05	Performance	Report generation time	Large dataset exists	Generate report and measure	Large date range	Report generated within 3 seconds.	Pending	Pending

NFR-03 – Reliability

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
UT_NFR-03_01	Reliability	Record persistence	Database available	Save record; retrieve it	Valid record	Record remains unchanged.	Pending	Pending
IT_NFR-03_02	Reliability	Backup creation	Backup process configured	Run/schedule backup	Attendance database	Backup is created successfully.	Pending	Pending
IT_NFR-03_03	Reliability	Restore backup	Valid backup exists	Restore backup to test database	Backup file	Records are restored correctly.	Pending	Pending
SYS_NFR-03_04	Reliability	Refresh reliability	Record saved	Refresh and reopen record	Saved record	No data loss or duplicate.	Pending	Pending
SYS_NFR-03_05	Reliability	Failure recovery	Temporary service failure simulated	Retry after failure	Valid attendance data	System recovers without corrupting records.	Pending	Pending

3. Test Execution Instructions

- Execute every test case manually using the steps and test data provided.
- Enter the observed outcome in the Actual Result column.
- Mark Test Result as Pass when the actual result matches the expected result; otherwise mark Fail.
- Record defects for every failed test case and retest after correction.
- For NFR-02, record measured response time; for NFR-03, record backup and restore evidence.

4. Test Summary

Category	Number of Test Cases
Functional requirements	60
Security NFR-01	5
Performance NFR-02	5
Reliability NFR-03	5
Total	75